# StockSense Pro

## RFID-Based Smart Inventory Management System with Security & Usage Analytics

## Project Idea

StockSense Pro is a smart inventory management system developed using embedded system technologies to automate stock monitoring and improve warehouse security. The project combines RFID-based product identification, password-protected access control, real-time inventory tracking, low-stock alerts, and usage analytics into a single compact system.

The system was initially designed using Arduino Uno, but due to memory and storage limitations, the hardware platform was upgraded to ESP32 for improved performance and scalability. The project uses RFID cards to identify products, an RTC module for date and time logging, an OLED display for real-time information, and a keypad-based authentication system for secure access.

StockSense Pro reduces manual inventory errors, improves monitoring efficiency, and provides a low-cost smart inventory solution suitable for warehouses, retail stores, laboratories, and small businesses.

# 1. Introduction

Inventory management is an important part of modern businesses and warehouses. Traditional inventory systems often rely on manual counting and handwritten records, which can lead to human errors, stock mismatch, and security problems.

To solve these issues, StockSense Pro was developed as a smart RFID-based inventory management system. The system automatically tracks stock movement using RFID cards and provides secure access through password authentication. It also records stock-in and stock-out activities with date and time information using an RTC module.

The project focuses on automation, security, and analytics while keeping the overall system low-cost, user-friendly, and efficient.

# 2. Project Objectives

The main objectives of this project are:

- Automate inventory tracking using RFID technology
- Reduce human error in stock management
- Provide secure access using password authentication
- Monitor stock-in and stock-out operations
- Display real-time inventory information
- Generate basic inventory analytics
- Detect low-stock conditions automatically
- Maintain date and time logs for inventory activities
- Create a scalable system for future IoT integration

# 3. System Features

### Main Features of StockSense Pro

- RFID-based product identification
- Admin and staff login system
- Password-protected inventory access
- Real-time stock monitoring
- OLED display user interface
- Low-stock alert using buzzer and LED
- Inventory analytics and usage tracking
- Customer favorite / best-selling product detection
- Date and time logging using RTC module
- Auto stock-in functionality
- Multiple product management
- Interactive keypad menu system

# 4. Components Used

| Component | Quantity | Purpose |
|---|---|---|
| ESP32 Development Board | 1 | Main controller and processing unit |
| RC522 RFID Module | 1 | RFID card reading |
| RFID Cards/Tags | 4+ | Product identification |
| OLED Display (128x64) | 1 | System interface and information display |
| RTC DS3231 Module | 1 | Real-time date and time tracking |
| 4x4 Matrix Keypad | 1 | Password and menu input |
| Active Buzzer | 1 | Audio alerts |
| Yellow LED | 1 | Warning indication |
| Breadboard | 1 | Circuit prototyping |
| Jumper Wires | Multiple | Circuit connections |
| USB Power Supply | 1 | System power |
| SPI & I2C Communication | Built-in | Data communication |

# 5. Software and Libraries Used

| Software / Library | Purpose |
|---|---|
| Arduino IDE | Programming environment |
| MFRC522 Library | RFID communication |
| RTClib | RTC date and time management |
| Adafruit SSD1306 | OLED display control |
| Adafruit GFX | Graphics support |
| Keypad Library | Keypad input handling |
| Wire Library | I2C communication |
| SPI Library | SPI communication |

# 6. Working Principle

The system works in multiple stages:

## Stage 1 – System Initialization

When the system powers on:

- ESP32 initializes all modules
- OLED display shows startup screen
- RFID module and RTC module become active
- Login menu is displayed

## Stage 2 – User Authentication

The user selects:

- Admin Login
- Staff Login

A 4-digit password is entered through the keypad.

### Access Conditions

- Correct password → Access granted
- Wrong password → Buzzer alert and access denied

## Stage 3 – RFID Product Scanning

After successful login:

- RFID cards are scanned
- Each RFID UID matches a specific product
- Product details are identified automatically

Products Used:

- T-Shirt – RFID B1
- Pant – RFID B3
- Tracksuit – RFID  B4
- Shoe – White RFID

## Stage 4 – Stock Management

After product detection:

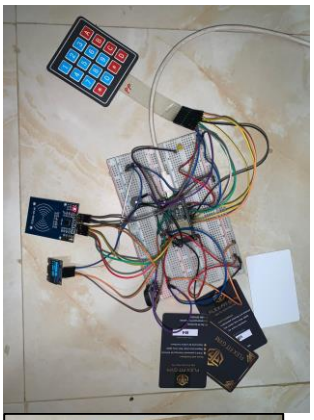
- Stock quantity updates automatically
- Sold counter increases
- Remaining stock is displayed
- Date and time are recorded

### Stage 5 – Analytics and Monitoring

The system provides:

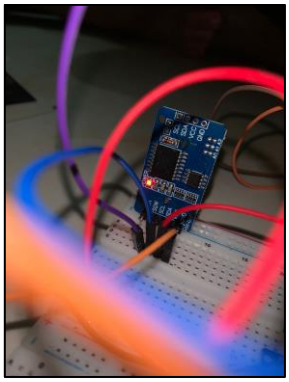
- Total stock report
- Total sold report
- Customer favorite product
- Low stock detection
- Inventory status analytics

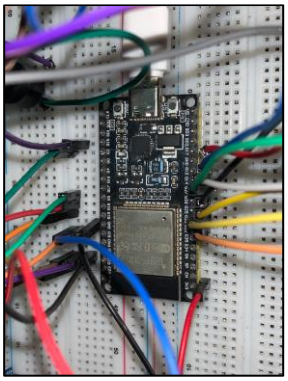
# 7. Project Showcasing

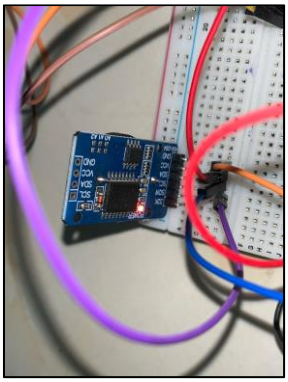


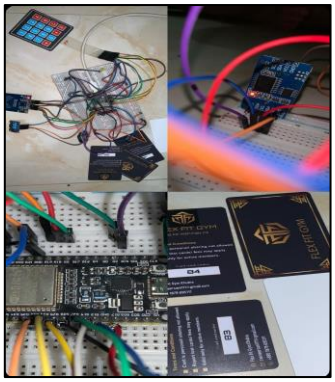












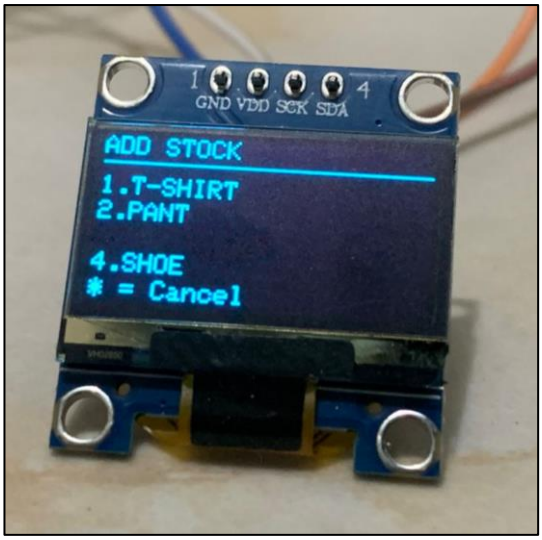

GND VDD SCK SDA
STOCKSENSE


GND VDD SCK SDA
WELCOME TO
WAREHOUSE
MANAGEMENT


GND VDD SCK SDA
STOCKSENSE PRO
A = ADMIN LOGIN
B = STAFF LOGIN


GND VDD SCK SDA
ADMIN ACCESS
GRANTED




# 8. Security System

StockSense Pro includes multiple security features:

- Password authentication for admin and staff
- Separate admin and staff access modes
- Unknown RFID card detection
- Audio and visual warning system
- Access denial for unauthorized users

These features help protect inventory from unauthorized access and misuse.

# 9. Inventory Analytics

The system includes basic usage analytics features such as:

- Total sold products
- Best-selling product detection
- Remaining stock count
- Inventory summary reports
- Real-time stock monitoring

This allows the user to identify frequently used products and manage restocking efficiently.

# 10. Limitations Faced During Development

## Initial Hardware Limitation

At the beginning of the project, Arduino Uno was used as the main controller. However, several limitations appeared during implementation:

### Problems with Arduino Uno

- Limited SRAM memory
- Limited flash storage
- Difficulty handling multiple libraries together
- OLED graphics consumed high memory
- RTC and RFID operations slowed performance
- Limited scalability for future upgrades

Because of these limitations, the project was migrated to ESP32.

# 11. Improvements After Using ESP32

After switching to ESP32:

- System performance improved significantly
- More memory became available
- Faster processing speed
- Better multitasking support
- Stable OLED display operation
- Improved RFID reading response
- Future IoT integration became possible

This upgrade solved most of the hardware limitations faced during early development.

# 12. Current Limitations

Although the project works successfully, some limitations still exist:

- Data is not permanently stored after power loss
- RFID scanning range is limited
- Internet/cloud connectivity is not implemented
- Limited number of RFID product entries
- No mobile application support

# 13. Future Improvements

Several future upgrades can make the system more powerful:

- Cloud database integration
- IoT-based remote monitoring
- Mobile app control
- Wi-Fi inventory dashboard
- Firebase or AWS support
- Fingerprint authentication
- Barcode scanner integration
- Email/SMS stock alerts
- AI-based demand prediction
- Automatic report generation

# 14. Applications

StockSense Pro can be used in:

- Retail shops
- Warehouses
- Laboratories
- Libraries
- Hospitals

- Smart stores
- Educational institutions

# 15. Conclusion

StockSense Pro successfully demonstrates how embedded systems and RFID technology can be used to automate inventory management in a smart and secure way.

The project combines authentication, RFID identification, stock tracking, real-time monitoring, analytics, and alert systems into one compact solution. The migration from Arduino Uno to ESP32 improved system stability, memory handling, and future scalability.

Overall, the project provides an efficient, low-cost, and user-friendly inventory management solution suitable for real-world applications and future IoT expansion.